

Project Nyra - Architecture & Implementation Whitepaper (v1)

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Stack decisions locked in this doc: Nexus Router (grafbase/nexus) + LiteLLM + OpenRouter, Dify + Activepieces + n8n, TwentyCRM as system-of-record CRM, standalone Nyra Admin UI (shadcn + Magic UI), Graphiti + Letta memory layer from day 0, and a scriptable Quote API.

> Not legal advice. This document includes compliance considerations and risk controls; run everything past qualified counsel before production use.

0) Executive summary

Project Nyra is an AI-augmented mortgage operations platform with:

- Borrower-facing logistics automation (status updates, doc requests, scheduling, message relay) with strict topic boundaries.
- Agent-facing orchestration for internal workflows (lead triage, campaign selection, quote preparation, CRM hygiene).
- A “memory cube”: Graphiti temporal knowledge graph + Letta memory manager, mirrored against TwentyCRM and enriched by conversation events.
- Campaign & Quote control plane in a standalone Nyra Admin webapp (shadcn + Magic UI), separate from TwentyCRM's UI.

The primary engineering goal is high reliability + compliance-first automation:

- deterministic workflows are executed by n8n (timers, drip campaigns, retries) + Activepieces (connectors + approvals)
- conversational intelligence and application UX live in Dify
- all tools & models are routed through Nexus Router, with LiteLLM handling provider routing and telemetry.

1) Goals & non-goals

Goals

1. Beat Bonzo/AgentLegend feature-for-feature, then add:
 - multi-system memory, graph intelligence, and workflow observability
 - quote automation with audit trails
 - campaign editing and preview with compliance guardrails
2. Borrower communications constrained to logistics and doc collection.
3. Single source of truth: TwentyCRM.
4. Production-grade governance: routing, auth, rate limits, audit logs.

Non-goals (for v1)

- Replacing TwentyCRM entirely (we extend it).
- Fully automated underwriting/loan pricing decisions (human approval required).
- Voice (Kokoro TTS) ships later - explicitly reserved in the design.

2) High-level system diagram

3) Why this stack (and what each piece owns)

Nexus Router (grafbase/nexus)

Nexus is the unified routing + governance plane for:

- MCP tool aggregation (one endpoint for many tools)
- LLM provider routing
- centralized auth + rate limits + observability

LiteLLM + OpenRouter

- LiteLLM: model routing, spend analytics, provider fallback.
- OpenRouter: multi-model access across providers.

Dify (embedded in Nyra Admin + borrower portals)

Dify owns:

- The chat UI and “AI app runtime”
- Knowledge for grounded answers (but we prefer Graphiti for relationships)
- Exposing Dify apps as MCP if needed later.

Activepieces

Activepieces owns:

- “hands” - app integrations, approvals, connector catalog.
- MCP tool surfaces that agents can call.

n8n

n8n owns:

- Long-running orchestration: drip campaigns, schedules, retries, timers
- Integration glue: webhooks from lead sources, syncing events.

TwentyCRM

TwentyCRM is system-of-record:

- leads, people, pipeline stages, activity.
- You build a custom UI that operates on top of Twenty via API.

Graphiti + Letta

- Graphiti: temporally-aware knowledge graph memory for agents.
- Letta: memory manager deciding what becomes durable memory.

4) Compliance & legal risk controls (high-level)

4.1 TCPA / robocall & text compliance

Maintain an auditable consent ledger, and manage opt-outs across all channels. FCC “one-to-one consent” changes (effective Jan 27, 2025) should be assumed relevant to lead-gen-style outreach.

4.2 CAN-SPAM

Email must include opt-out mechanisms and honor opt-out quickly.

4.3 GLBA Safeguards / privacy

If you handle consumer financial information, implement a written security program, access controls, encryption, incident response, vendor due diligence.

4.4 CFPB and chatbot risk

The CFPB has publicly stated it is monitoring chatbots in consumer finance and expects institutions to meet obligations and avoid harming consumers.

Nyra control strategy

- Borrower bot is “logistics-only”, with a classifier gate.

- Anything about rates, approvals, underwriting decisions → human escalation.
- Every outbound message: “You can opt out” + channel-specific compliance.
- Full audit trail: who said what, why, and what tool was used.

5) Core workflows (v1)

5.1 Lead intake

Sources:

- RateHunter form
- LendingTree / FreeRateUpdate feeds
- LeadMailbox (if integrable)

Flow:

1. Ingest → n8n webhook
2. Normalize lead → create Person + MortgageLead in Twenty
3. Pick campaign (rules engine) → schedule steps via n8n
4. Queue quote job → Quote API

5.2 Campaign execution

- Campaign authoring in Nyra Admin
- Campaign runner in n8n
- Channel delivery via Activepieces connectors (SMS/email/voicemail provider)
- Consent / opt-out enforced centrally.

5.3 Quote generation

- Quote API consumes scenario + borrower facts
- Pricing adapters (Rocket/LenderPrice/etc.) - initial phase may be manual import
- Output: multi-option quote JSON + (later) PDF; writeback to Twenty + Graphiti

5.4 Borrower conversation (logistics-only)

Dify app “Borrower Concierge” reads pipeline stage + missing docs, offers scheduling, logs events.

5.5 Internal operator assistant

Dify app “Nyra Ops” can draft messages, build campaigns, prepare quotes; approval required for outbound sends.

6) MCP toolchain (recommended minimum)

Route all tools through Nexus:

- GitHub MCP Server
- Filesystem MCP (dev only)
- Docker Hub MCP
- Docker MCP Gateway / Toolkit (optional dev convenience)
- Serena MCP (semantic code retrieval/editing)
- Archon MCP (project knowledge/tasks) - internal only

7) Repo layout & bootstrapping

This package includes a proposed structure:

- bootstrap/ - scripts to clone forks, configure env, run dev/prod, set up Gitea and CI
- vendor/ - cloned forks (claude-flow, archon)
- prod/vendor-images/ - Dockerfiles to build pinned production images from those forks

- infra/ - docker-compose stacks, monitoring, secrets patterns
- apps/ - nyra-admin + ratehunter
- services/ - quote-api + campaign-engine

References (URLs)

- Nexus Router docs: <https://nexusrouter.com/docs>
- Grafbase Nexus repo: <https://github.com/grafbase/nexus>
- Dify MCP docs: <https://docs.dify.ai/en/use-dify/publish/publish-mcp> and <https://docs.dify.ai/en/use-dify/build/mcp>
- Activepieces MCP overview: <https://www.activepieces.com/blog/model-context-protocol-mcp>
- Docker Hub MCP Server: <https://docs.docker.com/ai/mcp-catalog-and-toolkit/hub-mcp/>
- Docker MCP Gateway: <https://docs.docker.com/ai/mcp-catalog-and-toolkit/mcp-gateway/>
- GitHub MCP: <https://github.com/github/github-mcp-server> and <https://docs.github.com/en/copilot/how-tos/provide-context/use-mcp/use-the-github-mcp-server>
- Graphiti: <https://github.com/getzep/graphiti> and docs: <https://help.getzep.com/graphiti/getting-started/mcp-server>
- Letta memory mgmt: <https://docs.letta.com/advanced/memory-management/>
- CAN-SPAM guide (FTC): <https://www.ftc.gov/business-guidance/resources/can-spam-act-compliance-guide-business>
- GLBA overview (FTC): <https://www.ftc.gov/business-guidance/privacy-security/gramm-leach-bliley-act>
- CFPB chatbot post: <https://www.consumerfinance.gov/about-us/blog/cfpb-has-entered-the-chat/>